

# TrES-5b

R-0888

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_250819 · created 2026-07-29T01:37:13+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460906.8133 +/- 0.0043 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.103 +/- 0.054                 |
| Transit depth (Rp/Rs)^2                           | 1.06 +/- 1.11 [%]               |
| Orbital Inclination (inc)                         | 79.79 +/- 2.85                  |
| Airmass coefficient 1 (a1)                        | 0.46 +/- 0.37                   |
| Airmass coefficient 2 (a2)                        | 0.85 +/- 0.92                   |
| Scatter in the residuals of the lightcurve fit is | 103.08 %                        |
| Best Comparison Star                              | #2 - [271, 312]                 |
| Optimal Aperture                                  | 5.16                            |
| Optimal Annulus                                   | 8.17                            |
| Transit Duration (day)                            | 0.024 +/- 0.028                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                         |
|---------------------------|-----------------------------------------|
| Rp/R■ fit vs published    | 0.103 ± 0.054 vs 0.142 (deviation 0.4σ) |
| Duration fit vs published | 0.024 d vs 0.0758 d (deviation 1.03σ)   |
| Mid-time O–C              | -1.3 min                                |
| Depth SNR                 | 1.1                                     |
| Residual scatter          | 103.08 %                                |

### Light curve

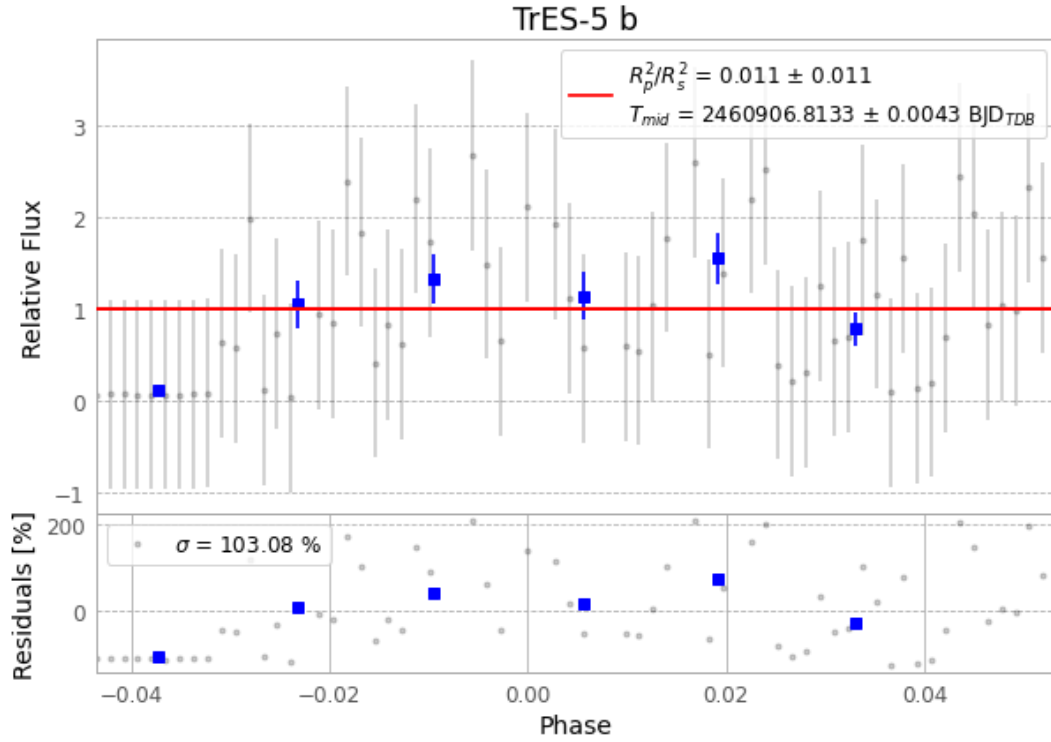


Figure 1 — FinalLightCurve\_TrES-5 b\_2025-08-18.png (SHA-256 fb0df1ed258b8625...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 278], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 969928427f16db73...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

70 FITS frames (46 MB), TRES-5250819055415.FITS → TRES-5250819092115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250819.log (0952d9c1c5f1...), FinalLightCurve\_TrES-5 b\_2025-08-18.png (fb0df1ed258b...), FinalLightCurve\_TrES-5 b\_2025-08-18.csv (6fc7769ea30e...)

## Compute resources

Wall time 185.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 01:37Z.